

Discrete Aggregate Domination in L^q -Spaces

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Status

This packet gives a candidate partial result, likely valid, for Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. It proves the discrete independent-increment aggregate-domination comparison in every L^q -space for $1 < p, q < \infty$.

It is partial relative to the full source problem because it does not yet package the full general-local-martingale extension, and it does not cover the endpoint $p = 1$.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source crop below states that the general-local-martingale case is open and reduces the missing accessible-jump issue to a discrete independent-increment comparison.

...now use η_j independence and η_j being in L^q for each j (2.10), hence from (12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n \geq 1}$ and $(\eta_n)_{n \geq 1}$ be X -valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C \mathbb{E} \left\| \sum_n \eta_n \right\|^p?$$

The next page adds that, by symmetrization, one may assume the variables are symmetric, and that even this symmetric discrete case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n 's and η_n 's are symmetric. But even the symmetric case is not known for the author.

Main Result

Theorem 1. *Let $1 < p, q < \infty$, let (S, Σ, ρ) be a measure space, and put $X = L^q(S)$. Let $(\xi_i)_{i=1}^n$ and $(\eta_i)_{i=1}^m$ be finite families of independent mean-zero X -valued random variables. Assume that*

$$\sum_i \mathbb{P}(\xi_i \in B) \leq \sum_j \mathbb{P}(\eta_j \in B) \quad (B \in \mathcal{B}(X), 0 \notin B).$$

Then

$$\mathbb{E} \left\| \sum_i \xi_i \right\|_{L^q(S)}^p \leq C_{p,q} \mathbb{E} \left\| \sum_j \eta_j \right\|_{L^q(S)}^p.$$

Corollary 1. *Under the assumptions of Theorem 1,*

$$\mathbb{E} \max_{1 \leq k \leq n} \left\| \sum_{i=1}^k \xi_i \right\|_{L^q(S)}^p \leq C_{p,q} \mathbb{E} \max_{1 \leq k \leq m} \left\| \sum_{j=1}^k \eta_j \right\|_{L^q(S)}^p.$$

Idea of the Proof

The previous ℓ_q packet worked at $p = q$ because the q -moment of the lattice square function is additive across coordinates. Off the natural exponent, that additivity disappears. The correct replacement is the sharp L^q -valued Rosenthal theorem of Dirksen and Yaroslavtsev [2].

For independent mean-zero L^q -valued variables z_i , their theorem says that

$$\left(\mathbb{E} \left\| \sum_i z_i \right\|_{L^q}^p \right)^{1/p} \simeq_{p,q} \| (z_i) \|_{s_{p,q}},$$

where $s_{p,q}$ is built, by sums and intersections, from three gauges:

$$S_q(z) = \left\| \left(\sum_i \mathbb{E} |z_i|^2 \right)^{1/2} \right\|_{L^q}, \quad D_{q,q}(z) = \left(\sum_i \mathbb{E} \|z_i\|_{L^q}^q \right)^{1/q}, \quad D_{p,q}(z) = \left(\sum_i \mathbb{E} \|z_i\|_{L^q}^p \right)^{1/p}.$$

These gauges depend only on the aggregate occurrence measure $\sum_i \mathcal{L}(z_i)$.

The only point that is not completely formal is the presence of sum spaces, such as $S_q + D_{q,q}$. A near-optimal decomposition of the dominating variables induces a measure on split triples $(x, y_1, \dots, y_r)$, where $x = \sum_\ell y_\ell$. If the aggregate law of (ξ_i) is dominated by that of (η_j) , we restrict this split measure by the Radon-Nikodym derivative of the smaller aggregate law. This transports the decomposition to the ξ_i 's and decreases every basic gauge. Thus the full Rosenthal norm is monotone under aggregate domination.

The Rosenthal Norms

Let $X = L^q(S)$, $1 < p, q < \infty$. For a finite sequence $z = (z_i)$ of X -valued random variables define

$$S_q(z) = \left\| \left(\sum_i \mathbb{E} |z_i|^2 \right)^{1/2} \right\|_{L^q(S)},$$

$$D_{r,q}(z) = \left(\sum_i \mathbb{E} \|z_i\|_{L^q(S)}^r \right)^{1/r}, \quad r \in \{p, q\}.$$

The Dirksen-Yaroslavtsev independent Rosenthal theorem [2, Theorem 2.1] says that for independent mean-zero z_i ,

$$\left(\mathbb{E} \left\| \sum_i z_i \right\|_{L^q(S)}^p \right)^{1/p} \simeq_{p,q} \|z\|_{s_{p,q}},$$

where $s_{p,q}$ is one of the following six sum/intersection spaces:

$$\begin{aligned} S_q \cap D_{q,q} \cap D_{p,q}, & \quad 2 \leq q \leq p < \infty, \\ S_q \cap (D_{q,q} + D_{p,q}), & \quad 2 \leq p \leq q < \infty, \\ (S_q \cap D_{q,q}) + D_{p,q}, & \quad 1 < p < 2 \leq q < \infty, \\ (S_q + D_{q,q}) \cap D_{p,q}, & \quad 1 < q < 2 \leq p < \infty, \\ S_q + (D_{q,q} \cap D_{p,q}), & \quad 1 < q \leq p \leq 2, \\ S_q + D_{q,q} + D_{p,q}, & \quad 1 < p \leq q \leq 2. \end{aligned}$$

Intersections carry the maximum norm and sums carry the usual infimum norm.

Aggregate Monotonicity

For a finite sequence $z = (z_i)$, let

$$\mu_z = \sum_i \mathcal{L}(z_i)|_{X \setminus \{0\}}$$

be its aggregate occurrence measure. The next lemma is the key point.

Lemma 1. *Let $a = (a_i)$ and $b = (b_j)$ be finite sequences of X -valued random variables with $\mu_a \leq \mu_b$. Then*

$$\|a\|_{s_{p,q}} \leq \|b\|_{s_{p,q}}.$$

Proof. First consider the three basic gauges. Since $\mu_a \leq \mu_b$,

$$\int_X |x|^2 d\mu_a(x) \leq \int_X |x|^2 d\mu_b(x)$$

pointwise on S , and therefore $S_q(a) \leq S_q(b)$. Similarly,

$$D_{r,q}(a)^r = \int_X \|x\|_{L^q}^r d\mu_a(x) \leq \int_X \|x\|_{L^q}^r d\mu_b(x) = D_{r,q}(b)^r$$

for $r = p, q$.

This proves the assertion for every expression involving only intersections. It remains to handle sums. We prove the required monotonicity recursively for any expression built from the three basic gauges using $\cap$ and $+$. The intersection step is immediate from the maximum norm. For the sum step, write the expression as $E + F$. Fix $\varepsilon > 0$, and choose a decomposition $b = u + v$ such that

$$\|u\|_E + \|v\|_F \leq \|b\|_{E+F} + \varepsilon.$$

Here $u = (u_j)$, $v = (v_j)$, and $b_j = u_j + v_j$.

The decomposition induces a finite measure λ on

$$\{(x, y, z) \in X^3 : x = y + z\}$$

by

$$\lambda = \sum_j \mathcal{L}(b_j, u_j, v_j)|_{X \setminus \{0\} \times X \times X}.$$

Its x -marginal is μ_b . Let $h = d\mu_a/d\mu_b$, $0 \leq h \leq 1$, and replace λ by $h(x)\lambda(dx, dy, dz)$. Disintegrating this restricted measure over its x -marginal μ_a , and enlarging the probability spaces of the a_i 's if necessary, we obtain random variables a_i^E, a_i^F such that

$$a_i = a_i^E + a_i^F$$

and the aggregate occurrence measures of (a_i^E) and (a_i^F) are dominated by those of u and v , respectively. By the induction hypothesis,

$$\|(a_i^E)\|_E \leq \|u\|_E, \quad \|(a_i^F)\|_F \leq \|v\|_F.$$

Thus

$$\|a\|_{E+F} \leq \|(a_i^E)\|_E + \|(a_i^F)\|_F \leq \|b\|_{E+F} + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ completes the recursive proof.

The above disintegration is first applied on the separable closed subspace generated by the essentially separable ranges of the finite family involved; standard approximation then removes boundedness or simplicity restrictions. $\square$

Proof of the Main Result

Let $\mu_\xi = \sum_i \mathcal{L}(\xi_i)|_{X \setminus \{0\}}$ and $\mu_\eta = \sum_j \mathcal{L}(\eta_j)|_{X \setminus \{0\}}$. The hypothesis is exactly $\mu_\xi \leq \mu_\eta$. By Lemma 1,

$$\|(\xi_i)\|_{s_{p,q}} \leq \|(\eta_j)\|_{s_{p,q}}.$$

The independent L^q -valued Rosenthal theorem gives

$$\begin{aligned} \left(\mathbb{E} \left\| \sum_i \xi_i \right\|_{L^q}^p \right)^{1/p} &\lesssim_{p,q} \|(\xi_i)\|_{s_{p,q}} \\ &\leq \|(\eta_j)\|_{s_{p,q}} \\ &\lesssim_{p,q} \left(\mathbb{E} \left\| \sum_j \eta_j \right\|_{L^q}^p \right)^{1/p}. \end{aligned}$$

Raising to the p -th power proves Theorem 1.

For Corollary 1, apply Doob's L^p maximal inequality to the martingale $\sum_{i \leq k} \xi_i$, use Theorem 1, and then bound the terminal η -sum by the η -maximum.

Verification and Limitations

The proof uses the lower and upper bounds in the Dirksen-Yaroslavtsev Rosenthal theorem, so it is restricted to $1 < p < \infty$. Their paper records some endpoint upper bounds, but the two-sided norm identification needed here is not available in this packet at $p = 1$.

This packet proves the discrete independent-increment bottleneck from Remark 12.10 in L^q -spaces. A full general-local-martingale L^q theorem should now be within reach, but it requires a separate audit combining this jump comparison with continuous Gaussian covariance domination and the source paper's decoupled-tangent reduction. The present packet deliberately does not claim that final step.

The code file `code/grouping_simulation.py` is only a finite sanity check. It compares several groupings of the same aggregate coordinate jump law in ℓ_q^N . The tested ratios remain dimension-stable, supporting the monotonicity proof but not serving as evidence in place of it.

Novelty Check

Cheap run indexes contained the previous c_0 counterexample, Hilbert positive packets, and the ℓ_q natural-exponent packet, but no L^q all- $p > 1$ aggregate-domination packet for arXiv:1907.11588. Web searches for exact phrases around **characteristically dominated**, ℓ_q , L^q , and **aggregate domination** did not locate an existing statement of this consequence. The decisive external input is the Dirksen-Yaroslavtsev L^q -valued Burkholder-Rosenthal theorem [2]; the monotonicity lemma above is the additional identification that connects that theorem to Remark 12.10.

Human Review Recommendation

Review Lemma 1 first, especially the transfer of near-optimal sum-space decompositions by disintegration and Radon-Nikodym restriction. If that lemma is accepted, the rest of the proof is an immediate application of the cited Rosenthal theorem.

The next target is to combine this packet with the full decoupled-tangent reduction in [1]. The jump part should follow from the present theorem; the continuous part should follow from Gaussian covariance domination, but it should be audited separately before promoting a full L^q general-local-martingale result.

References

- [1] I. S. Yaroslavtsev, *Local characteristics and tangency of vector-valued martingales*, arXiv:1907.11588.
- [2] S. Dirksen and I. S. Yaroslavtsev, *L^q -valued Burkholder-Rosenthal inequalities and sharp estimates for stochastic integrals*, arXiv:1707.00109.